

DIN: Distributed Inference Network

- Scenarios, Requirements, and Experiments

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DeepSeek Opens the Era of Inclusive AI with Explosive Growth in Inference Services

DeepSeek-R1 achieves accuracy comparable to OpenAI's o1 model with 70% lower computation requirements, becoming a global focus.

Market Success : Surpassed ChatGPT in downloads across **140** app markets; **fastest** application to exceed **30 million** daily active users (DAUs)

AI models are evolving from chat tools to production and daily life tools, forming irreversible new business scenarios, bringing huge opportunities for model distribution and edge inference

- Humans and AI
- API and AI
- AI and AI (agents and agents, workflows and workflows)

A new distributed inference architecture to handle 100 millions of users and 10 millions of concurrent access

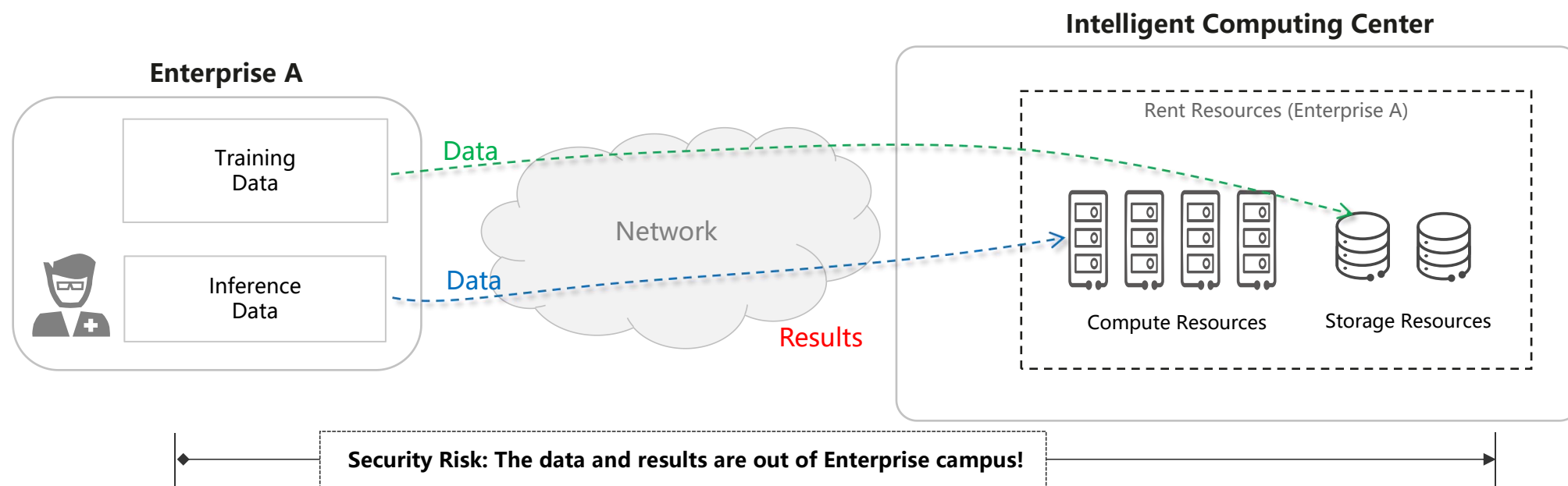
DeepSeek " *The server is busy. Please try again later* " : has experienced multiple outages due to surging access volumes, frequently becoming unavailable. This highlights the inability of the centralized deployment architecture to handle massive concurrent user access.

Comparison with Gemini: with a similar DAU scale, uses a **distributed architecture**, allowing users to access the nearest nodes with much less network latency, without access issues.

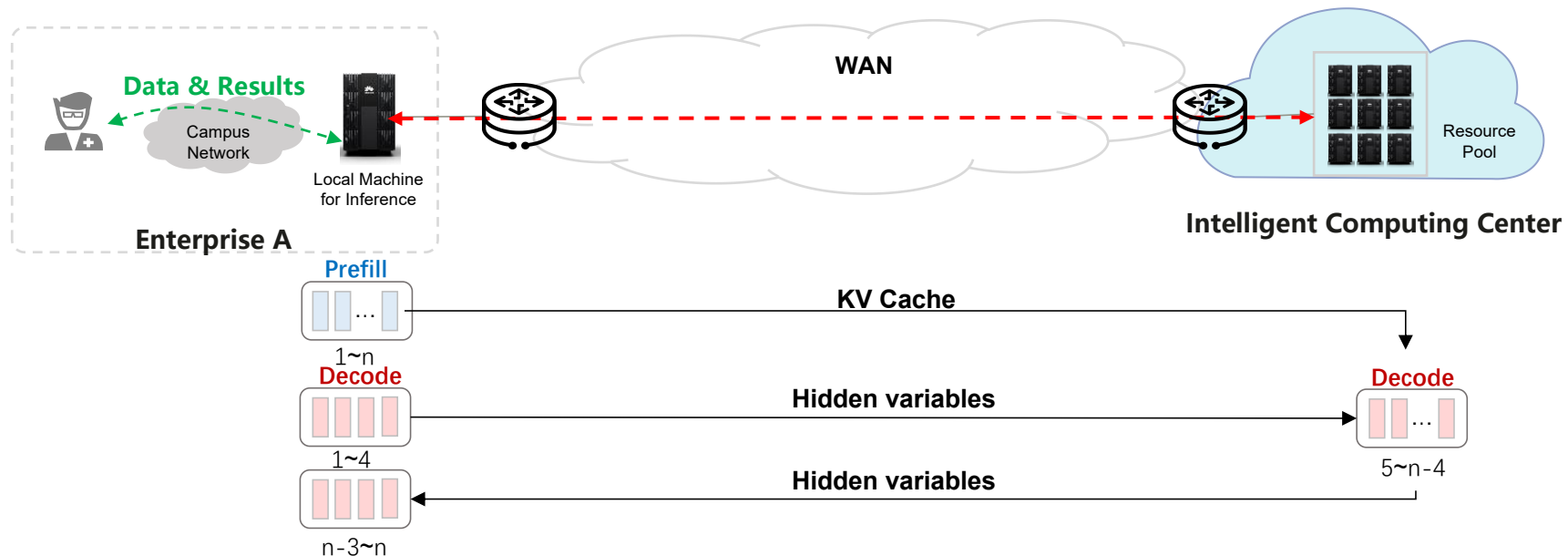
■ Carrier Network Opportunity:

- **Internet era:** CDN distributed web/video content from centralized pools to edge nodes, solving congestion and large-scale user access
- **Inclusive AI era:** DIN (**Distributed Inference Network**) uses a *distributed* inference architecture to provide ubiquitous low-latency inference services

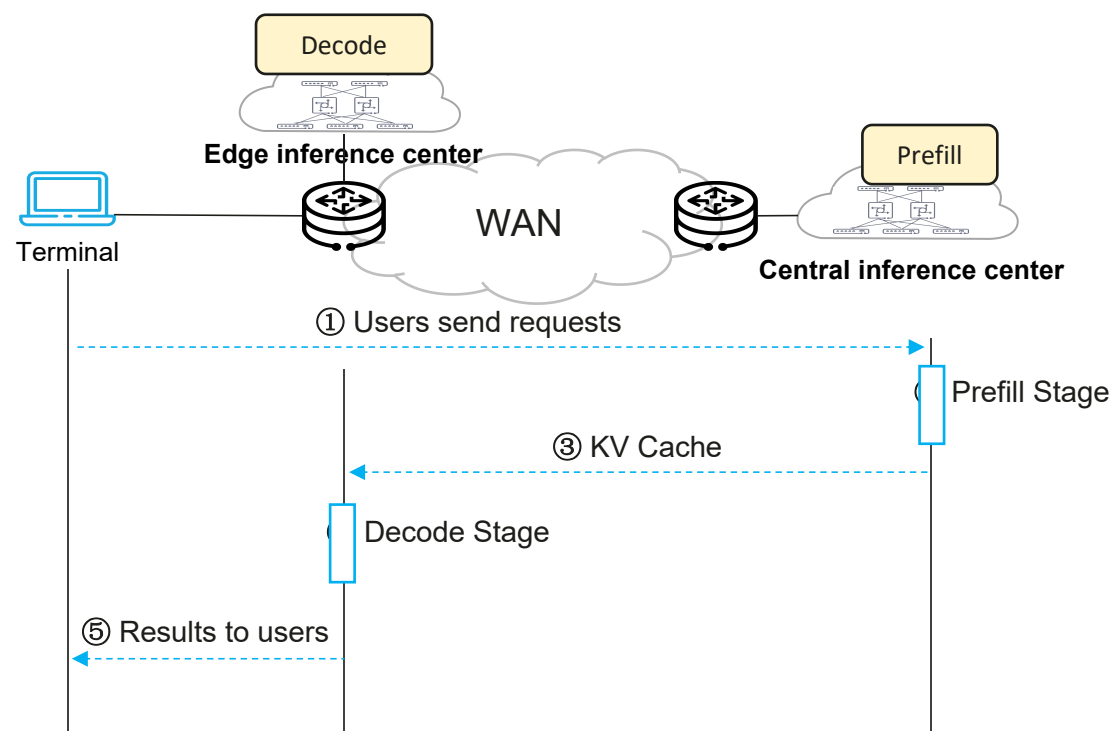
- Enterprises desire to have high quality inference models (e.g. DeepSeek) to help their fast growth.
- **However, to build local inference infrastructure is very challenging:**
 - Frequent scale-up
 - Long construction period
 - High construction cost due to Liquid cooling and power
- So it is desirable to **rent** for the resources in the Cloud
- **However, the Enterprises' data and results will have to go out of their campus, which is not acceptable.**



- How to keep the Enterprises' data and results inside their campus while renting the resources in Cloud?
 - One of the available techniques is **Split Learning**
 - It **divides a model into parts, allowing** different parts **residing on different devices (clients and servers)**
 - Enhancing privacy and reducing resource requirements for certain devices
- Split Learning can be jointly used with **Prefill-Decode (PD) Separation** technique
 - PD separation is a technique in large language model (LLM) serving that optimizes performance by **dividing the inference process into distinct prefill and decode stages**, often **running** these stages **on separate hardware or software components**.
 - Some parts of the Decode stage is moved to the intelligent computing center, reducing the resource requirements in Enterprises.
- **The network needs to guarantee the ultra-low latency, ultra-low jitter and zero loss to support the real-time and consistent inference.**



- For providing delay sensitive services at home such as Embodied Intelligence, both PD stages are deployed in the Cloud but in different places -> **edge and central inference centers need to coordinate**
 - The Prefill is deployed in the central inference center – The Prefill is compute resource sensitive
 - The Decode is deployed in the edge inference center – The Decode is storage resource sensitive
- KV Cache is generally MB to GB, so in order to guarantee the first token' s time requirement, the KV Cache needs to be transmitted within 100ms, meaning **the WAN needs to provide steady high bandwidth.**
- RDMA is used between P and D, the impact of loss is severe, meaning **the WAN needs to provide ultra-low loss.**



Experiments on PD separation

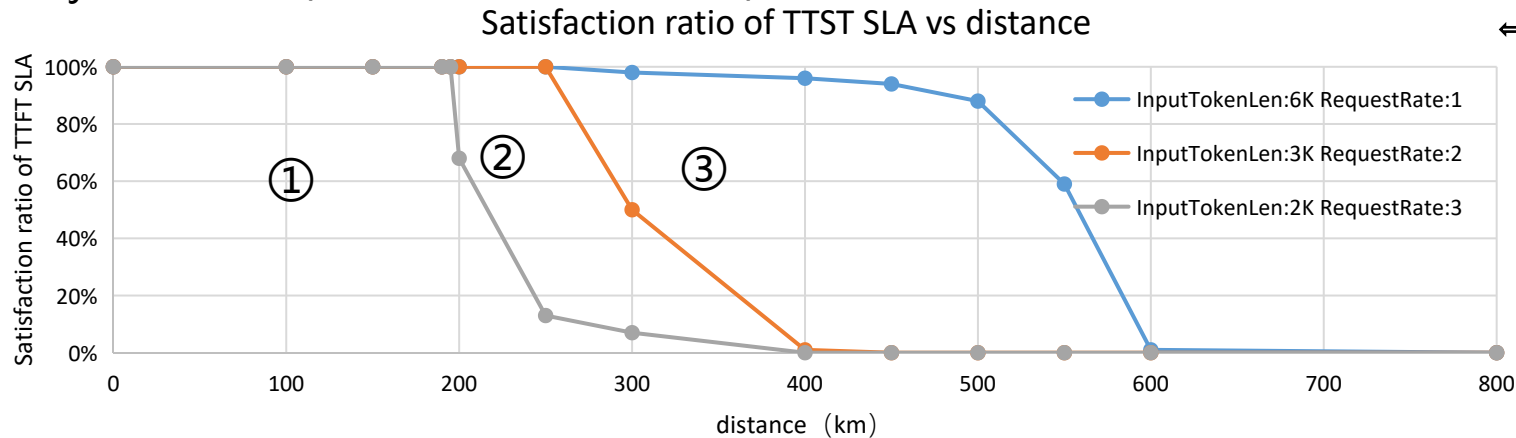
Although the disaggregated PD deployment has become the de facto standard in modern LLM serving systems, it remains unclear whether it is feasible to further increase the **geographical distance** of disaggregated PD deployment.

Experiments setup

- Using two Ascend servers (8 NPU with 200 Gbps RDMA NIC each)
- PD-disaggregated LLaMA3-70B inference system
- MindIE over RoCE

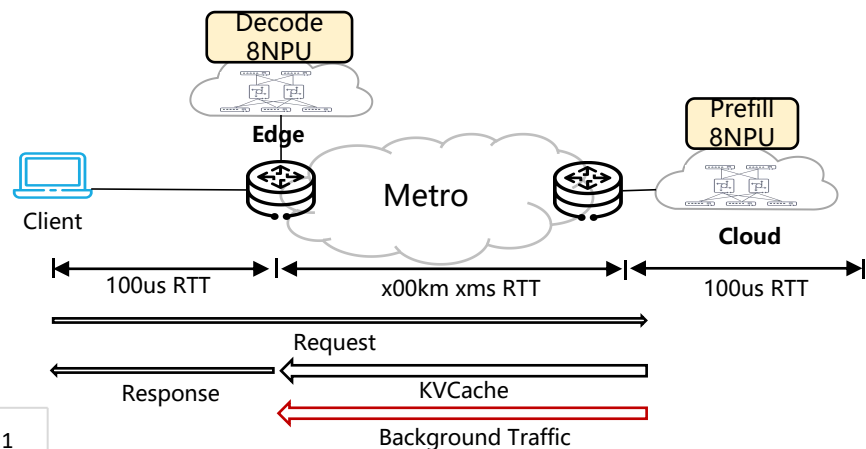
Results

- In Fujian Province(P8+D8, 910B4, MindIE)

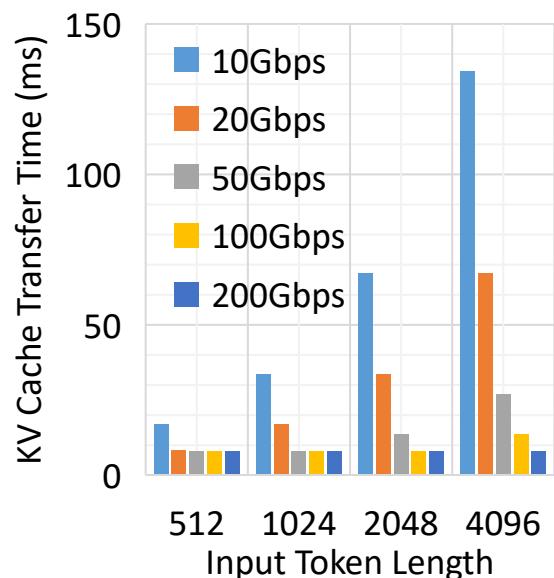


- PD separation deployment (200-300 km) shows no performance degradation, based on the current MindIE
- PD separation beyond 200-300 km introduces performance degradation challenges

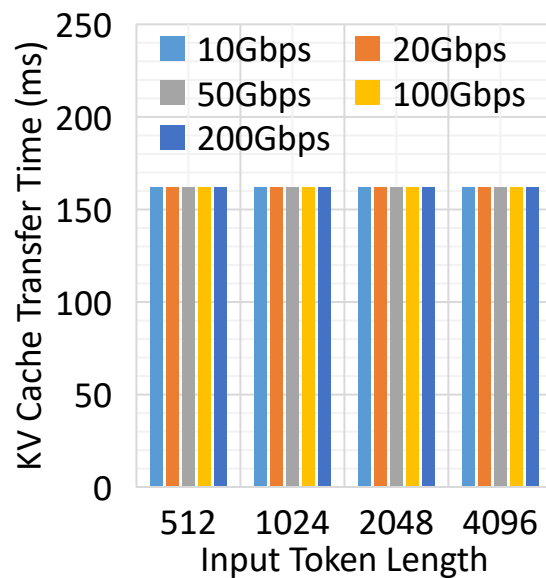
Accepted in Sigcomm workshop on networks for AI computing (NAIC)



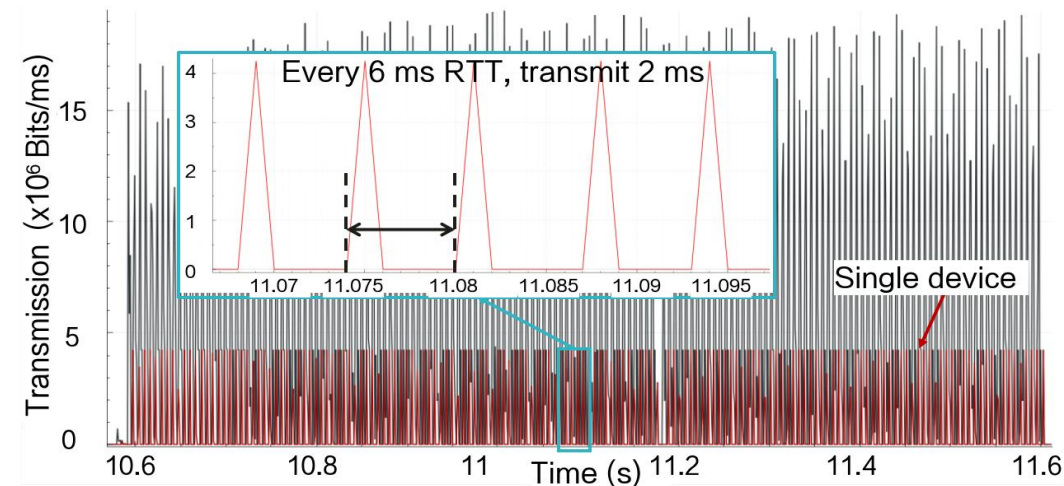
- A brute-force increase in bandwidth in distributed P/D deployments fails to improve TTST
- Every 6 ms RTT, transmit 2 ms, leaving network bandwidth idle during RTT waiting periods



Centralized Deployment
RTT=0.05ms



Distributed Deployment
RTT=1ms



Transmission pattern of KV cache

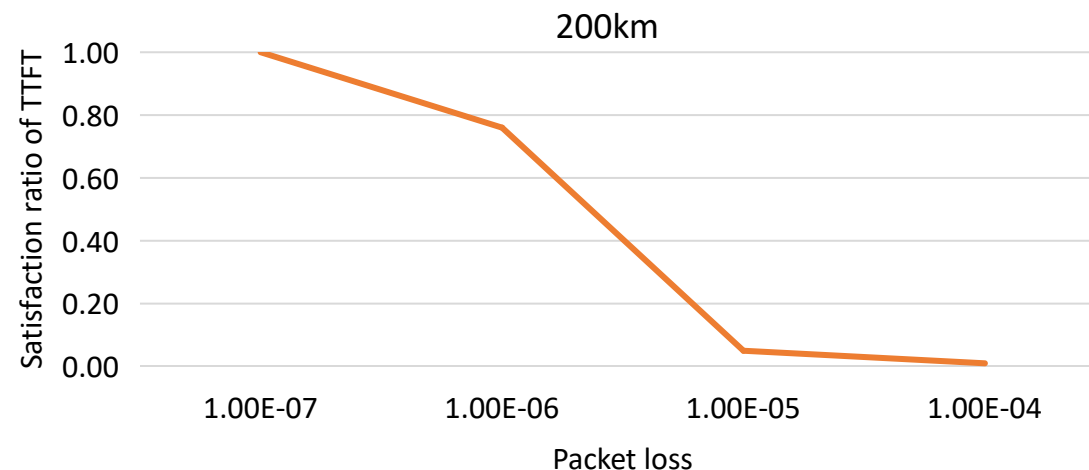
- RTT-bound instead of bandwidth-bound
- optimize the transmission protocol

- 0.1% packet loss causes a 90% decrease in throughput.

Damage instrument setting and the throughput

Distance/KM	delay (ms)	RTT (ms)	packet loss	100GE rate
100 KM	0.5ms	1ms	1e-6	82.7Gbps
100 KM	0.5ms	1ms	1e-5	58.6Gbps
100 KM	0.5ms	1ms	1e-4	22.5Gbps
100 KM	0.5ms	1ms	1e-3	9.1Gbps
100 KM	0.5ms	1ms	1e-2	1.3Gbps

- At a distance of 200 km, PD separation requires a packet loss ratio of less than $1e-7$.



- The packet loss rate of the current computing power network is $1e-5$ to $1e-6$, not meeting the requirements for high computing efficiency.

- ToB Inference Service

- The WAN needs to guarantee the ultra-low latency, ultra-low jitter and zero loss to support the real-time and consistent inference.

- ToH Inference Service

- The WAN needs to provide steady high bandwidth.
- The WAN needs to provide ultra-low loss.



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Thank You for Listening!

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